

```
import kagglehub

# Download latest version
path = kagglehub.dataset_download("kritikseth/fruit-and-vegetable-image-recognition")

print("Path to dataset files:", path)

↗ Path to dataset files: /kaggle/input/fruit-and-vegetable-image-recognition

import os

os.listdir(path)
folder_train = os.path.join(path, "train")
os.listdir(folder_train)
```

```
↗ ['capsicum',
  'sweetcorn',
  'orange',
  'tomato',
  'turnip',
  'ginger',
  'raddish',
  'pomegranate',
  'pineapple',
  'jalepeno',
  'apple',
  'carrot',
  'lettuce',
  'bell pepper',
  'eggplant',
  'beetroot',
  'kiwi',
  'pear',
  'cabbage',
  'cauliflower',
  'paprika',
  'lemon',
  'sweetpotato',
  'grapes',
  'cucumber',
  'corn',
  'banana',
  'garlic',
  'chilli pepper',
  'watermelon',
  'mango',
  'peas',
  'onion',
  'potato',
  'spinach',
  'soy beans']
```

```
import matplotlib.pyplot as plt

from PIL import Image
from collections import defaultdict

image_size = defaultdict(int)
train_folder = os.path.join(path, "train")
# img_path = os.path.join(train_folder, os.listdir(train_folder)[0])
# os.listdir(img_path)
for label in os.listdir(train_folder):
    # print(os.listdir(folder))
    for file in os.listdir(train_folder):
        list_file = os.listdir(os.path.join(train_folder, file))
        for img_path in list_file:
            finalPath = os.path.join(train_folder, file, img_path)
            img = Image.open(finalPath)
            # print(train_folder, file, img_pathr)

            # build_filepath = os.path.join(train_folder, file, img_pathr)
            try:
                image_size[img.size] += 1
```

```

except:
    pass
for size,count in image_size.items():
    print(f'{size} , {count}')
```



```

-----
KeyboardInterrupt                                Traceback (most recent call last)
<ipython-input-3-56ed65810093> in <cell line: 0>()
    15         for img_path in list_file:
    16             finalPath = os.path.join(train_folder,file,img_path)
----> 17             img = Image.open(finalPath)
    18             # print(train_folder, file,img_pathr)
    19

----- 4 frames -----
/usr/local/lib/python3.11/dist-packages/PIL/JpegImagePlugin.py in _open(self)
    361         i = s[0]
    362         if i == 0xFF:
--> 363             s = s + self.fp.read(1)
    364             i = i16(s)
    365         else:

KeyboardInterrupt:
```

Check ends of file

```

from tensorflow.keras.preprocessing.image import load_img, img_to_array, array_to_img
import tensorflow as tf
from google.colab import drive
```

```

myDrive = drive.mount('/content/drive')
save_path = '/content/drive/MyDrive/ml/Giuaky/train_resized'
source_path = os.path.join(path, "train")
target_size = (256,256)
# {'jpg', 'png', 'jpeg'}
for label_folder in os.listdir(source_path):
    print(label_folder)
    pathlabel = os.listdir(os.path.join(source_path,label_folder))
    output_folder = os.path.join(save_path,label_folder)
    os.makedirs(output_folder, exist_ok=True)
    for img_file in pathlabel:
        pathImage = os.path.join(os.path.join(source_path,label_folder),img_file)
        print(pathImage)
        try:
            if img_file.lower().endswith(('jpg','png','jpeg')):
                img = load_img(pathImage, target_size=target_size)
                print(f'test {img}')
                save_path_full = os.path.join(output_folder,img_file)
                img.save(save_path_full)
                print(f'save {save_path_full}')
            except Exception as e:
                print(f"Error with image {pathImage}: {e}")
```





```
test <PIL.Image.Image image mode=RGB size=256x256 at 0x7988B0430E50>
save /content/drive/MyDrive/ml/GiuaKy/train_resized/soy beans/Image_20.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_64.jpg
test <PIL.Image.Image image mode=RGB size=256x256 at 0x7988BD4314D0>
save /content/drive/MyDrive/ml/GiuaKy/train_resized/soy beans/Image_64.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_52.jpg
test <PIL.Image.Image image mode=RGB size=256x256 at 0x7988BE53CD0>
save /content/drive/MyDrive/ml/GiuaKy/train_resized/soy beans/Image_52.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_24.jpg
test <PIL.Image.Image image mode=RGB size=256x256 at 0x79896079C190>
save /content/drive/MyDrive/ml/GiuaKy/train_resized/soy beans/Image_24.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_73.jpg
test <PIL.Image.Image image mode=RGB size=256x256 at 0x7988BDAE3210>
save /content/drive/MyDrive/ml/GiuaKy/train_resized/soy beans/Image_73.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_76.jpg
test <PIL.Image.Image image mode=RGB size=256x256 at 0x7988CEC2E590>
save /content/drive/MyDrive/ml/GiuaKy/train_resized/soy beans/Image_76.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_29.jpg
test <PIL.Image.Image image mode=RGB size=256x256 at 0x7988BE53CD0>
save /content/drive/MyDrive/ml/GiuaKy/train_resized/soy beans/Image_29.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_57.jpg
test <PIL.Image.Image image mode=RGB size=256x256 at 0x7988BEA9E390>
save /content/drive/MyDrive/ml/GiuaKy/train_resized/soy beans/Image_57.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_37.jpg
test <PIL.Image.Image image mode=RGB size=256x256 at 0x798960880950>
save /content/drive/MyDrive/ml/GiuaKy/train_resized/soy beans/Image_37.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_6.jpg
test <PIL.Image.Image image mode=RGB size=256x256 at 0x7988CCF4AC10>
save /content/drive/MyDrive/ml/GiuaKy/train_resized/soy beans/Image_6.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_9.jpg
test <PIL.Image.Image image mode=RGB size=256x256 at 0x798960880950>
save /content/drive/MyDrive/ml/GiuaKy/train_resized/soy beans/Image_9.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_71.jpg
test <PIL.Image.Image image mode=RGB size=256x256 at 0x7988BD463C90>
save /content/drive/MyDrive/ml/GiuaKy/train_resized/soy beans/Image_71.jpg
```

Test Transform and resized with dataset

```
from tensorflow.keras.preprocessing.image import load_img, img_to_array, array_to_img
import tensorflow as tf
from google.colab import drive
```

```
myDrive = drive.mount('/content/drive')
save_path_transform = '/content/drive/MyDrive/ml/GiuaKy/train_resized_transform'
source_path = os.path.join(path, "train")
target_size = (256,256)
```

```
def transform_image(image):
    image = tf.image.random_flip_left_right(image)
    image = tf.image.random_flip_up_down(image)
    image = tf.image.random_brightness(image, max_delta=0.2)
    image = tf.image.random_contrast(image, lower=0.8, upper=1.2)
    return image

for label_folder in os.listdir(source_path):
    print(label_folder)
    pathlabel = os.listdir(os.path.join(source_path, label_folder))
    output_folder = os.path.join(save_path_transform, label_folder)
    os.makedirs(output_folder, exist_ok=True)
    for img_file in pathlabel:
        pathImage = os.path.join(os.path.join(source_path, label_folder), img_file)
        print(pathImage)
        try:
            if img_file.lower().endswith(('jpg', 'png', 'jpeg')):
                img = load_img(pathImage, target_size=target_size)
                img_array = img_to_array(img) / 255
                img_tensor = tf.convert_to_tensor(img_array, dtype=tf.float32)

                transformed_image = transform_image(img_tensor)
                transformed_image = array_to_img(transformed_image)

                save_path_full = os.path.join(output_folder, img_file)
                transformed_image.save(save_path_full)
        except Exception as e:
            print(f"Error with image {pathImage}: {e}")
```

```

/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_41.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_10.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_81.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_47.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_30.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_77.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_59.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_23.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_60.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_63.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_94.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_50.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_70.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_83.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_32.png
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_98.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_21.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_95.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_93.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_25.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_17.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_87.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_15.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_42.png
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_54.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_90.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_1.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_89.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_43.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_40.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_91.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_97.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_34.png
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_56.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_55.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_78.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_35.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_80.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_8.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_84.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_38.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_67.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_46.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_49.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_3.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_100.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_20.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_64.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_52.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_24.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_73.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_76.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_29.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_57.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_37.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_6.jpg
/kaggle/input/fruit-and-vegetable-image-recognition/train/soy beans/Image_9.jpg
/kaggle/inout/fruit-and-veetable-image-recognition/train/sov beans/Image_71.jpg

```

```

print(source_path)
extensions = set()
for root, dirs, files in os.walk(source_path):
    for file in files:
        if '.' in file:
            extension = file.split('.')[1].lower()
            extensions.add(extension)

```

```
print(extensions)
```

```

/kaggle/input/fruit-and-vegetable-image-recognition/train
{'jpg', 'png', 'jpeg'}

```

Check size and total after resize

Number of images per class

```
labelfolder = os.listdir(save_path)
```

```

def count_images_in_folder(folder_path):
    counts = defaultdict(int)

```

```

counts = defaultdict(int)
for label in os.listdir(folder_path):
    label_path = os.path.join(folder_path, label)
    if os.path.isdir(label_path):
        counts[label] = len(os.listdir(label_path))
return counts
before = count_images_in_folder(source_path)
print(before)
after = count_images_in_folder(save_path)
print(after)

```



```

{'': 79, 'pineapple': 99, 'jalepeno': 88, 'apple': 68, 'carrot': 82, 'lettuce': 97, 'bell pepper': 90, 'eggplant': 84, 'beetroot': 88, 'k
': 90, 'eggplant': 84, 'beetroot': 88, 'kiwi': 88, 'cabbage': 92, 'cauliflower': 79, 'paprika': 83, 'lemon': 82, 'sweetpotato': 69, 'gra

```

Compare image dimensions

```

def get_image_size(folder_path):
    sizes = defaultdict(int)
    for label in os.listdir(folder_path):
        label_path = os.path.join(folder_path, label)
        if os.path.isdir(label_path):
            for img_file in os.listdir(label_path):
                img_path = os.path.join(label_path, img_file)
                img = Image.open(img_path)
                sizes[img.size] += 1
    return sizes
print(source_path)
print(save_path)
before = get_image_size(source_path)
print(sum(before.values()))

after = get_image_size(save_path)
print(sum(after.values()))
save_path_transform = '/content/drive/MyDrive/ml/GiuaKy/train_resized_transform'

save_path_check = get_image_size(save_path_transform)
print(sum(save_path_check.values()))

/kaggle/input/fruit-and-vegetable-image-recognition/train
/content/drive/MyDrive/ml/GiuaKy/train_resized_transform
3115
3115
3115

```

Split into train/val/set

Bắt đầu lập trình hoặc [tạo](#) mã bằng trí tuệ nhân tạo (AI).